

PVsyst - Simulation report

Standalone system

Project: standalonw with shading 24-02-2026

Variant: New simulation variant

Standalone system with batteries

System power: 410 kWp

George Town - India

**PVsyst V7.4.7**

VC0, Simulation date:
25/03/26 21:05
with V7.4.7

Project summary**Geographical Site****George Town**

India

Situation

Latitude 13.09 °N

Longitude 80.28 °E

Altitude 24 m

Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

George Town

Meteonorm 8.1 (1996-2015) - Synthetic

System summary**Standalone system****PV Field Orientation**

Fixed plane

Tilt/Azimuth 15 / 20 °

Standalone system with batteries**Near Shadings**

Linear shadings : Fast (table)

User's needs

Fixed constant load

75.0 kW

Global

657 MWh/Year

System information**PV Array**

Nb. of modules

630 units

Pnom total

410 kWp

Battery pack

Technology

Lead-acid, sealed, Gel

Nb. of units

3540 units

Voltage

240 V

Capacity

35400 Ah

Results summary

Useful energy from solar 590652 kWh/year

Specific production 1442 kWh/kWp/year

Perf. Ratio PR 76.48 %

Missing Energy 66348 kWh/year

Available solar energy 613087 kWh/year

Solar Fraction SF 89.90 %

Excess (unused) 7225 kWh/year

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General parameters

Standalone system

PV Field Orientation

Orientation

Fixed plane
Tilt/Azimuth 15 / 20 °

Near Shadings

Linear shadings : Fast (table)

Standalone system with batteries

Sheds configuration

Nb. of sheds 66 units

Sizes

Sheds spacing 5.65 m
Collector width 2.63 m
Ground Cov. Ratio (GCR) 46.5 %
Top inactive band 0.02 m
Bottom inactive band 0.02 m

Shading limit angle

Limit profile angle 12.5 °

User's needs

Fixed constant load
75.0 kW
Global
657 MWh/Year

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

PV Array Characteristics

PV module

Manufacturer AE Solar
Model AE 650ME-132
(Original PVsyst database)
Unit Nom. Power 650 Wp
Number of PV modules 630 units
Nominal (STC) 410 kWp
Modules 35 string x 18 In series

At operating cond. (50°C)

Pmpp 375 kWp
U mpp 617 V
I mpp 607 A

Controller

Universal controller
Technology MPPT converter
Temp coeff. -5.0 mV/°C/Elem.

Converter

Maxi and EURO efficiencies 97.0 / 95.0 %

Total PV power

Nominal (STC) 410 kWp
Total 630 modules
Module area 1957 m²
Cell area 1834 m²

Battery

Manufacturer Narada
Model AcmeG 12V 200
Technology Lead-acid, sealed, Gel
Nb. of units 177 in parallel x 20 in series
Discharging min. SOC 20.0 %
Stored energy 7005.6 kWh

Battery Pack Characteristics

Voltage 240 V
Nominal Capacity 35400 Ah (C10)
Temperature Fixed 20 °C

Battery Management control

Threshold commands as SOC calculation
Charging SOC = 0.90 / 0.75
approx. 265.8 / 253.4 V
Discharging SOC = 0.20 / 0.45
approx. 236.4 / 245.7 V



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Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 17 mΩ
Loss Fraction 1.5 % at STC

Serie Diode Loss

Voltage drop 0.7 V
Loss Fraction 0.1 % at STC

Module Quality Loss

Loss Fraction -0.8 %

Module mismatch losses

Loss Fraction 2.0 % at MPP

Strings Mismatch loss

Loss Fraction 0.2 %

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.962	0.892	0.816	0.681	0.440	0.000

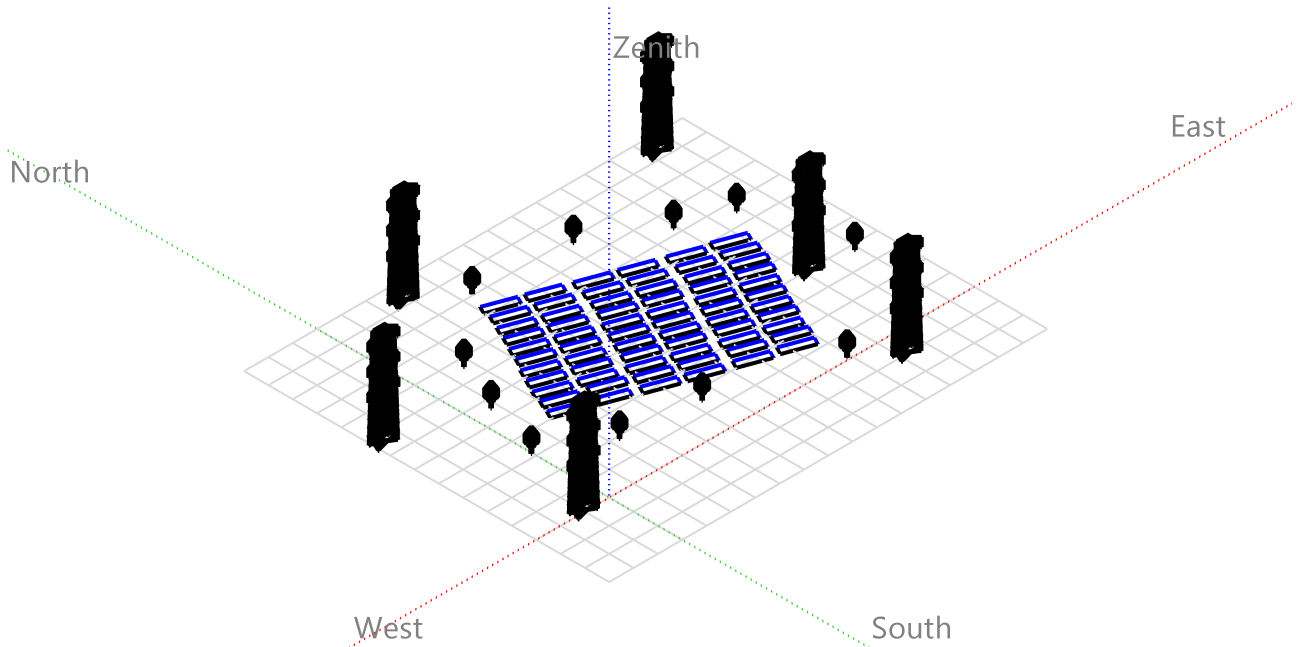


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Near shadings parameter

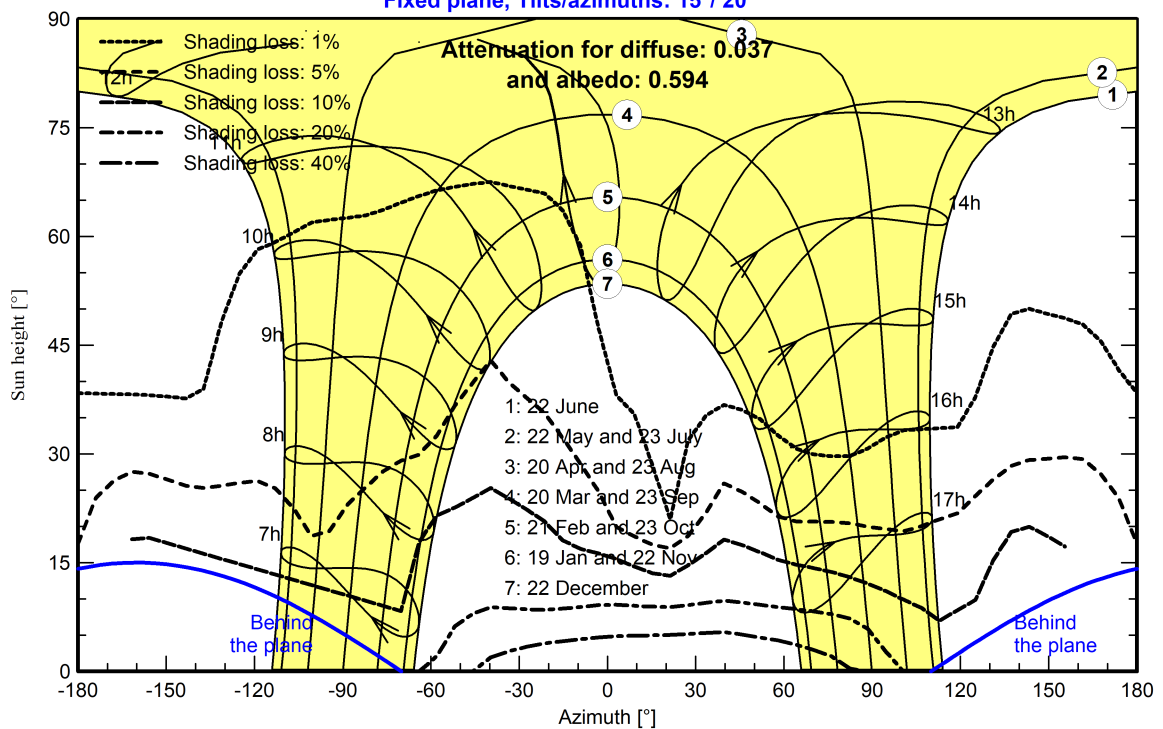
Perspective of the PV-field and surrounding shading scene



Iso-shadings diagram

Orientation #1

Fixed plane, Tilts/azimuths: 15°/ 20°





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Main results

System Production

Useful energy from solar 590652 kWh/year
Available solar energy 613087 kWh/year
Excess (unused) 7225 kWh/year

Loss of Load

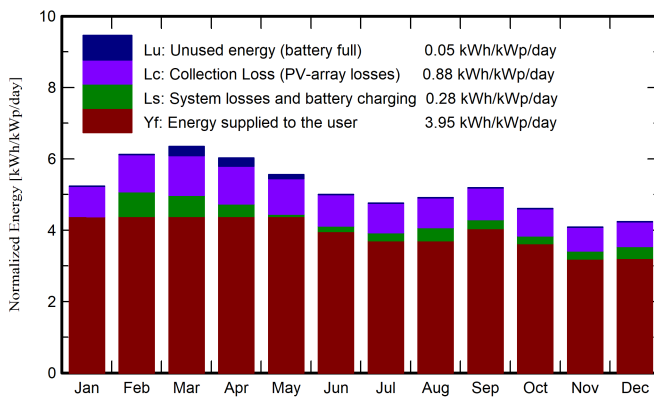
Time Fraction 0.0 %
Missing Energy 66348 kWh/year

Perf. Ratio PR 76.48 %
Solar Fraction SF 89.90 %

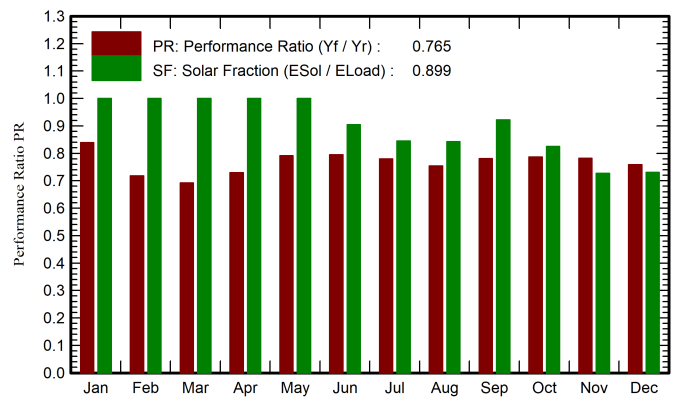
Battery aging (State of Wear)

Cycles SOW 96.5 %
Static SOW 91.7 %
Battery lifetime 12.0 years

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	GlobEff	E_Avail	EUnused	E_Miss	E_User	E_Load	SolFrac
	kWh/m ²	kWh/m ²	kWh	kWh	kWh	kWh	kWh	ratio
January	144.2	154.4	53347	0	0	55800	55800	1.000
February	156.4	164.2	55916	0	0	50400	50400	1.000
March	189.7	189.4	63868	3155	0	55800	55800	1.000
April	184.7	173.9	58578	2723	0	54000	54000	1.000
May	182.6	165.3	55522	1332	0	55800	55800	1.000
June	162.3	143.5	48408	2	5182	48818	54000	0.904
July	158.0	140.8	47775	4	8663	47137	55800	0.845
August	157.1	146.0	49438	0	8764	47036	55800	0.843
September	153.7	149.2	50445	0	4222	49778	54000	0.922
October	134.9	136.9	46570	0	9767	46033	55800	0.825
November	112.4	116.8	40122	2	14745	39255	54000	0.727
December	117.8	124.6	43097	7	15003	40797	55800	0.731
Year	1853.9	1804.8	613087	7225	66348	590652	657000	0.899

Legends

GlobHor Global horizontal irradiation
GlobEff Effective Global, corr. for IAM and shadings
E_Avail Available Solar Energy
EUnused Unused energy (battery full)
E_Miss Missing energy

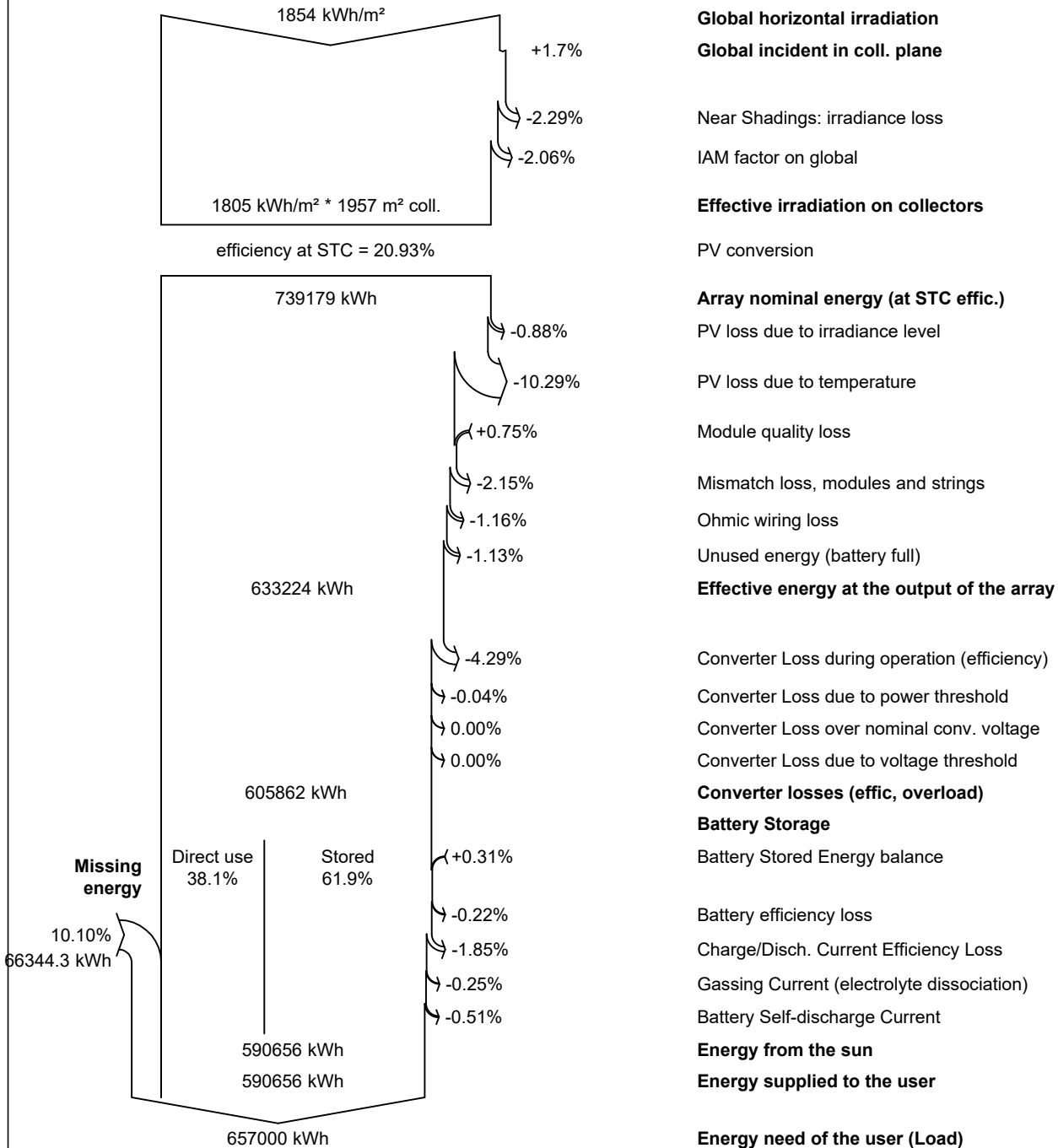
E_User Energy supplied to the user
E_Load Energy need of the user (Load)
SolFrac Solar fraction (EUsed / ELoad)



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Loss diagram





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Predef. graphs

Daily Input/Output diagram

